

Application scientist in AI-driven molecular discovery and materials science research with proven ability to translate complex scientific insights into strategic product development and AI solutions, driving innovation in drug discovery and advanced materials.

WORK EXPERIENCE

Assistant Professor, DCHEM, SRMUS, Sikkim 2025 –
- Teacher and training
Researcher
Mentor

QDX, Singapore (remote) – Computational chemist 2023 – 2024
- Refined existing and developed new AI-enhanced computational modelling protocols using open-source and proprietary tools, supporting internal R&D and two external drug discovery collaborations within one year.
- Delivered computational chemistry analyses for external partners, interpreting collaborator results and refining data to inform key drug discovery decisions and project direction.

Product Development:

- Served as subject matter expert alongside software engineers to design and build RUSH, a quantum mechanics & AI-based drug design platform. Guided feature development and scientific validation from concept through successful beta launch, bridging domain knowledge with engineering.
- Co-organized an internal AI hackathon; led data strategy by curating and standardizing diverse chemical and biological datasets, enabling teams to rapidly prototype new machine learning models.

JoVE, Bangalore – Science Editor 2021 – 2023
- Led and mentored a team of scientific writers to create visually engaging, accurate content (video and text) for publication, significantly improving content quality and audience engagement.
- Trained new science writers on specialized computational research software and scientific writing best practices, accelerating onboarding and fostering cross-team technical competency.
- Coordinated with cross-functional departments (video production, editorial, and marketing) to ensure on-time delivery of JoVE video articles, aligning scientific accuracy with product release schedules and organizational goals.

Kolabtree, Freelance Computational Chemistry Consultant

- Provided independent computational chemistry consulting for an antibody–drug conjugate design project. Developed molecular models and in-silico analyses to optimize drug-linker properties, contributing to the client's candidate selection and project success.

EDUCATION

PG certificate, Software engineering	IIT-Hyderabad, 2020-2021
Ph.D., Computational Chemistry	SSSIHL, Puttaparthi, 2015-2019
- SRF-Direct Fellow, CSIR	
- Sakura Science Exchange Fellow, JAIST, Japan	
M.Phil. Computational Chemistry, 8.2/10.0	SSSIHL, Puttaparthi, 2013-2015
M.Sc., Chemistry, GPA 4.86/5.0	SSSIHL, Puttaparthi, 2010-2012

RESEARCH EXPERIENCE

18 Publications [[google scholar](#), Citations: 370*, h-index 10, i10 index - 10]

- Primary research:
 - o AI integrated drug design, Drug repurposing, Metabolomics, Biosensors, Biomaterials
- Collaborative research:
 - o SPCE, Microfluidics, Sensors, Solar cells, Ion-conductive polymers

TECHNICAL SKILLS

- **Programming & Machine Learning:** Python (NumPy, pandas, SciPy, scikit-learn, PyTorch); proficient in data analysis, visualization, and algorithm development; familiar with python.
- **Computational Chemistry & Materials:** Molecular modelling (docking, molecular dynamics simulations), quantum chemistry (DFT and ab initio calculations), cheminformatics (RDKit, QSAR modelling, virtual screening); experience in materials simulation (polymer electrolytes).
- **Tools & Methodology:** Linux environment, HPC cluster computing, Git version control, strong cross-functional collaboration and project management skills.

TRAINING AND WORKSHOPS

- ChatGPT and generative AI 101 (2023), Growth school, Bangalore
- Molecular simulation with ML (2020), Princeton University
- International Winter School on Materials Science (2018), JNCASR, Bangalore
- Genomics, Proteomics, Drug Design and HPC (2017), SCF-Bio Facility, IIT-Delhi
- Winter School on 'Computational Chemistry' (2015), UGC-NRC, Hyderabad

REFERENCES

- Loong Wang, Founder – QDX, loong@qdx.co
- Dr Helene Kuhn, Manager – JOVE, helenekuhn86@gmail.com
- Dr. Jeff Layton, Freelance collaborator - Kolabtree, vleyton@uottawa.ca
- Dr. B Siva Kumar, PhD Supervisor, sivakumarbelliraj@gmail.com